1. An event has spacetime coordiantes (t,x) =(2.0 μs,1200 m) in reference frame S. What are the event's spacetime coordinates (a) in reference frame S$'$ that moves in the positive x-direction at 0.85c, and (b) in reference frame S$''$ that moves in the negative x-direction at 0.95c?

2. In the earth's reference frame, a tree is at the origin and a pole is at x = 30 km. Lightening strikes both the tree and the pole at t = 10 μs. The lightening strikes are observed by a rocket traveling in the x-direction at 0.50c.

A. What are the spacetime coordinates for these two events in the rocket's reference frame?

B. Are the events simultaneous in the rocket's frame? If not, which occurs first?

C. Determine Δs^2 for the two events.

3. A rocket cruising past earth at 0.80c, shoots a missile out the back door opposite the rockets motion, at 0.9c relative to the rocket. This is event A. According to the ship, the missile travels for 3 lm before hitting a target. This is event B.

A. Find the spacetime coordinates of Events A and B in the ship's frame.

B. Find the spacetime coordinates of Events A and B in the Earth's frame.

C. Find the spacetime coordinates of Events A and B in the Missile's frame.

4. A distant quasar is found to be moving away from the earth at 0.80c. A galaxy closer to earth and along the same line of sight is moving away from us at 0.20c. What is the recessional speed of the quasar, as a farction of c, as measure by astronomers in the galaxy?

5. An armada of spaceships that is 1.00 ly long (as measured in its rest frame) moves with speed 0.850c relative to Earth. A messenger travels from the rear of the armada to the front with a speed of 0.950 relative to Earth. How long does the trip take as measured (a) in the rest frame of the messenger, (b) in the rest frame of the armada, and (c) by an observer at Earth?

6. Suppose you want to use Mathematica to multiply the Matrix A = $\left(\begin{smallmatrix} 3 & 4 \\ -2 & 1 \end{smallmatrix}\right)$ by vector x =(3,1)

In math, the operation looks like

$$\begin{bmatrix} 3 & 4 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

To do this in mathematica type

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A = {{3,4},{-2,1}};
x={3,1};
A.x
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A. Use mathematica to write a script that computes the Lorentz transformation given some β.

B. According to frame S, event A happens at the spacetime coordinate (t,x) =(-5,10). Frame S$'$ passes frame S at t=0 going in the +x direction at $\beta = 0.999$. What are the spacetime coordinates of event A in frame S$'$?